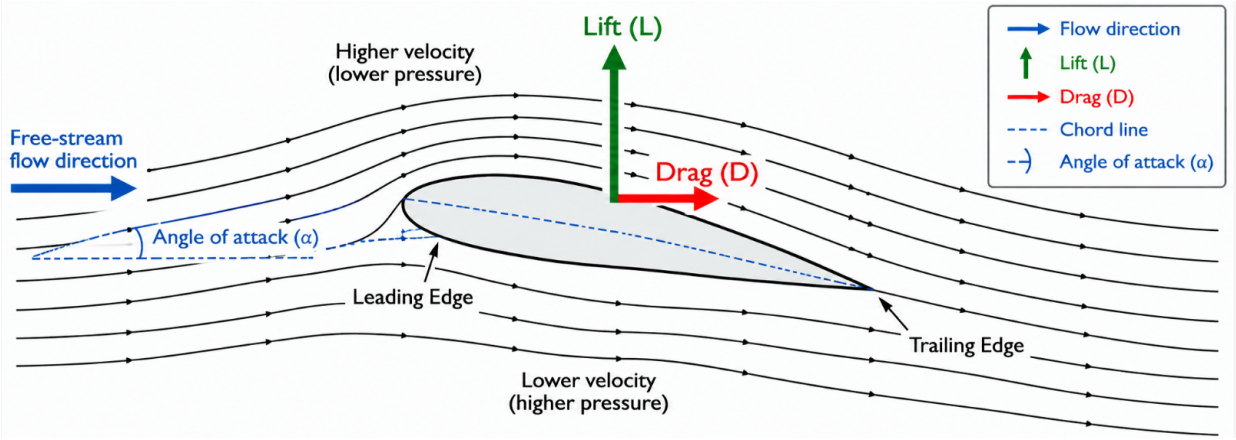
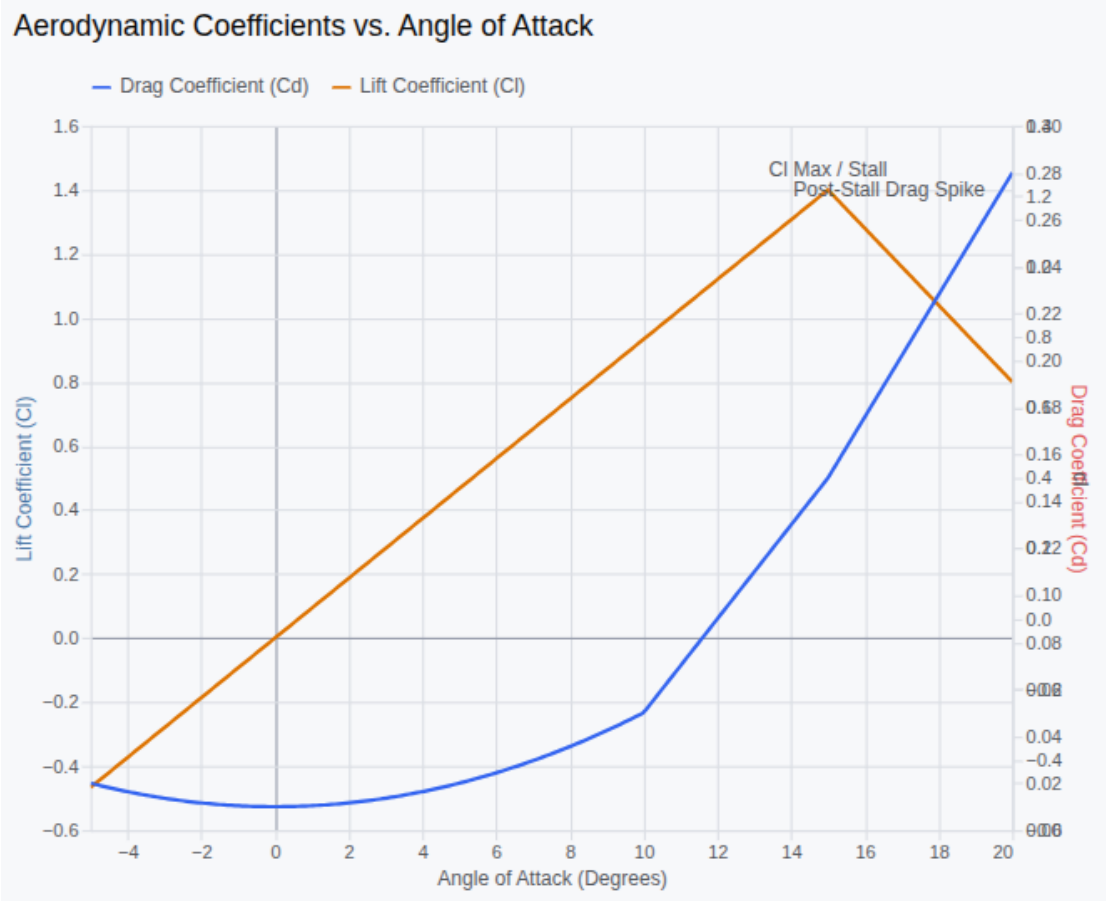
Supplemental examples

1. Estimate lift drag forces for a airfoil (cross sectional shape of a wing),
   given air density 1.225 kg/m^3, flow speed 10 m/s, area of this airfoil (in 2D t's a chord length)=1,
   and at AOA=6 degree, curves of coefficients vs AOA is provided





Hint:

$$L = \frac{1}{2}\rho v^2 A C_l = 1/2 * 1.225 * 10^2 * 1 * 0.6$$

Given a model rocket (Estes Vapor):
Body Tube Diameter: 1.64 inches (41.656 mm or 0.041656 m)

Nose Cone Length: 5.5 inches (Ogive shape)

Body Tube Length: 39.7 inches (Total length = 5.5 + 39.7 = 45.2 inches)

Fins: counts 3 fins, tip chord 2.0 inches, root chord 4.5 inches, semispan (sticks out from the body) 2.1 inches, and the front of the fin root starts 40 inches down from the nose tip

2. Each components' drag coefficients are estimated, calculate the total drag coefficients and the drag force when flying a speed of 20m/s at sea level.

Rocket Component	Drag coefficients
Nose Cone	0.040
Body Tube	0.339
Fin Set (3 Fins)	0.088
Base Drag	0.120
Interference	0.030

$$C_{d,\text{total}} = C_{d,\text{nose}} + C_{d,\text{body}} + C_{d,\text{fins}} + C_{d,\text{base}} + C_{d,\text{interference}}$$
$$= 0.040 + 0.339 + 0.088 + 0.120 + 0.030 = 0.617$$

$$D = \frac{1}{2}\rho v^2 A C_d = 0.5 * 1.225 * 20^2 * 3.14 * (0.041656/2)^2 * 0.617 = *N$$

3. Each components' normal force coefficients are estimated, calculate where is the Rocket Center of Pressure (CP)

components	Normal force Coefficients	locations (of own geometry, in inches)
Nose (ogive) cone	2.0	2.56
Body tube	0.0	unknow
Fin set (3 total)	9.7	1.66

Hint:
$$X_{CP} = \frac{(C_{N,nose} \cdot X_{nose}) + (C_{N,fins} \cdot X_{fins})}{C_{N,nose} + C_{N,fins}} = \frac{(2.0 * 2.56) + (9.92 * (40 + 1.66))}{(2.0 + 9.92)}$$